

Maximum Subarray Sum

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to find the maximum sum of values in a contiguous, nonempty subarray.

Input

The first input line has an integer n : the size of the array.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Output

Print one integer: the maximum subarray sum.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x_i \leq 10^9$

Example

Input	Output
8 -1 3 -2 5 3 -5 2 2	9